Methodology Summary

The project followed a structured machine learning workflow to predict customer conversion behavior based on demographic information, browsing activity, engagement metrics, traffic sources, and purchase history.

1. Data Exploration and Analysis

The dataset was first analyzed using Exploratory Data Analysis (EDA) to understand feature distributions, identify missing values, detect outliers, and examine relationships between variables and the target class. Conversion rates across different user segments were studied to identify key behavioral patterns.

2. Data Preprocessing

Missing values in numerical features were handled using median imputation to maintain data integrity. High-cardinality features such as Campaign_Code were removed to avoid encoding complexity and overfitting. Categorical variables were transformed using Label Encoding and One-Hot Encoding to make them suitable for machine learning algorithms.

3. Feature Engineering

Additional features were created to capture user engagement and purchasing intent more effectively. These included Engagement_Score, Product_Interest_Ratio, Returning_Customer, Income_Time_Ratio, and High_Engagement. The objective was to provide the model with more informative behavioral indicators beyond the original dataset.

4. Model Development

Multiple classification algorithms were evaluated, including Logistic Regression, Random Forest, and CatBoost. An 80-20 stratified train-validation split was used to ensure balanced class representation during model evaluation.

5. Model Optimization

Class imbalance was addressed through class-weight balancing. The CatBoost model demonstrated the strongest performance and was further optimized using probability threshold tuning. Instead of the default threshold of 0.50, an optimal threshold of 0.34 was selected to maximize the F1 Score.

6. Evaluation and Final Model Selection

Models were compared using Accuracy, Precision, Recall, and F1 Score, with F1 Score chosen as the primary evaluation metric. The final CatBoost model achieved an F1 Score of approximately 0.565 and provided the best balance between identifying actual converters and minimizing false predictions.

This methodology ensured a systematic approach to data preparation, model development, optimization, and evaluation, resulting in a robust customer conversion prediction solution.

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#SummerAnalytics2026

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